Chang Xu

chang.xu@connect.ust.hk Hong Kong SAR

Education

PhD **The Hong Kong University of Science and Technology**, Physics — Clear Water Bay, Hong Kong SAR — Aug 2022 – Aug 2027
- Expected graduation: Aug. 2027

BSc **University of Science and Technology of China**, Applied Physics — Hefei, China — Aug 2018 – July 2022

Experience

Department of Physics, The Hong Kong University of Science and Technology, PhD Student — HKUST, HK SAR — Aug 2022 – present — 4 years
Working with Prof. Qin Xu to design the nonlinear mechanics of soft composite solids.

Department of Modern Physics, University of Science and Technology of China, Student Research Assistant — USTC, China — June 2020 – July 2022 — 2 years 2 months
Working with Prof. Bing-Hong Wang to investigate the asymmetric simple exclusion process on complex networks.

Presentations

CRF Joint Meeting on Active Matter at HKUST (Hong Kong SAR, 2025-11). Contributed talk: "Non-reciprocal active solids bringing shear jamming into soft material design".

2025 APS March Meeting (Anaheim, USA, 2025-03). Contributed talk: "Realizing mechanical non-reciprocity in soft composites through shear jamming".

2024 USTC-HKUST Joint Workshop on Condensed Matter Physics (Chaohu, China, 2024-12). Contributed talk: "Realizing mechanical non-reciprocity in soft composites through shear jamming".

2024 Frontiers in Non-equilibrium Physics (Kyoto, Japan, 2024-07). Flash presentation and Poster: "Designing nonlinear mechanics of soft composite solids using shear jamming".

2023 USTC-HKUST Joint Workshop on Condensed Matter Physics (Guangzhou, China, 2023-12). Poster: "Designing nonlinear mechanics of soft composite solids using shear jamming".

Skills

Design, construct and conduct mechanical experiments: CAD; 3D printing; nano-indentation; Raspberry Pi/Arduino automation; spin coating; metal/plastic processing with milling/drilling machine; bench lathe and other machine shop tools.

Imaging soft and complex systems using advanced techniques: Traction force microscopy, confocal microscopy, interferometry, and photoelasticimetry.

Digital image processing: Object identification, tracking, and stress field reconstructions.

Numerical simulation and modelling: Molecular Dynamics and Monte Carlo simulations.

Research Interests

Interests: Soft materials, Granular physics, Active matter, Biophysics, Biomechanics, Mechanobiology, Statistical mechanics, Nonlinear dynamics, Stochastic processes